Customer Churn Prediction

Data Science

March 20, 2026

Belle

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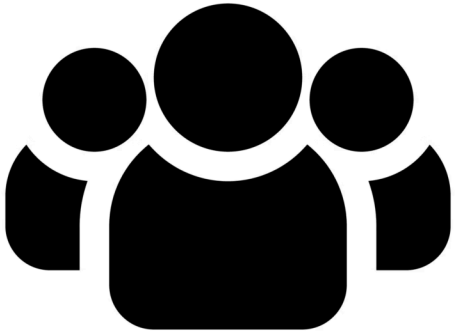
Problem Statement



Money Loss

PREDICTIVE ANALYTICS

Customers leave telecom companies

Loss of revenue

Need to predict churn

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Research Paper

From : www.ijcaonline.org

Churn Prediction using ML

An in-depth analysis of machine learning techniques applied to predictive churn modeling.

Telco Dataset Analysis

Leveraging comprehensive telecommunications datasets to uncover patterns in customer behavior.

Model Comparison

Evaluating and benchmarking various machine learning algorithms to identify the most accurate predictors.

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Understanding Our Data

Telco

Dataset Name

Telco Customer Churn Dataset

7,043

Total Customers

Individual records analyzed

21

Features

Dimensions per customer record

The dataset provides a robust foundation for predictive modeling, comprising 21 distinct customer attributes and a binary churn target variable for 7,043 unique observations.

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Feature Engineering & Target Variable

Dataset Features

Our dataset includes customer information such as tenure, monthly charges, contract type, and internet service options. These features help us understand customer behaviors and patterns.

Telcom Customer Churn

Each row represents a customer, each column contains customer's attributes described on the column Metadata.

The raw data contains 7043 rows (customers) and 21 columns (features).

The "Churn" column is our target.

customerID Customer ID	gender Whether the customer is a male or a female	SeniorCitizen Whether the customer is a senior citizen or not (1, 0)	Partner Whether the customer has a partner or not (Yes, No)	Dependents Whether the customer has dependents or not (Yes, No)
7043 unique values	Male 50% Female 50%	0 1	true 0 0% false 0 0%	true 0 0% false 0 0%
7590-VHVEG	Female	0	Yes	No
5575-GNVDE	Male	0	No	No
3668-QPYBK	Male	0	No	No
7795-CFOCW	Male	0	No	No

Target Variable

The target variable is **Churn**, a binary indicator showing whether a customer left (Yes) or stayed (No). This is what our model will predict.

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Data Preprocessing Pipeline

Cleaning and preparing data for machine learning models

01

Remove Irrelevant Columns

Eliminated customerID and other identifiers that don't contribute to predictions

02

Handle Missing Values

Identified and fixed missing data points using appropriate imputation methods

03

Encode Categorical Variables

Converted text-based categories into numerical format for model compatibility

04

Feature Scaling

Standardized numerical features to ensure equal weighting in model training

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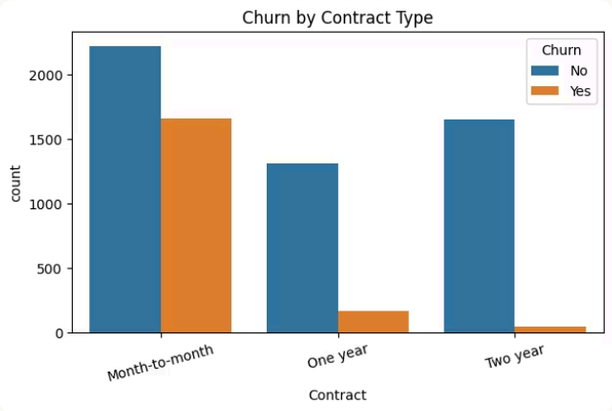
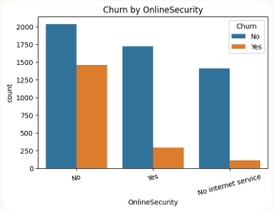
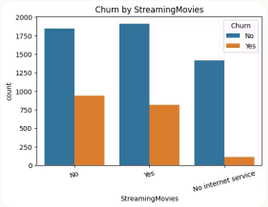
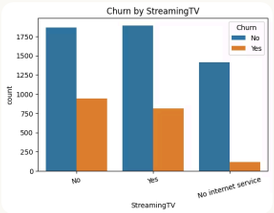
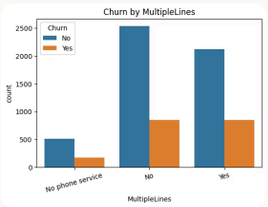
Exploratory Data Analysis Insights

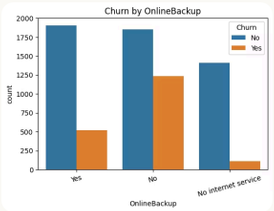


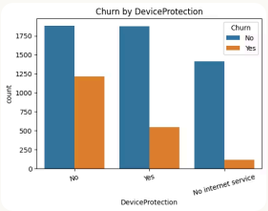


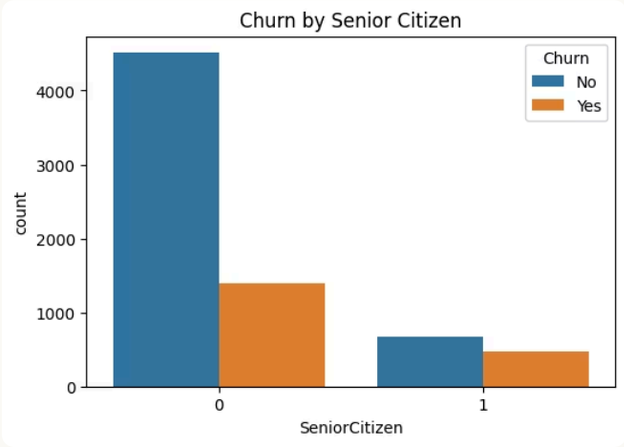












Contract Type Impact

Month-to-month contracts show significantly higher churn rates compared to annual or two-year agreements

Service Bundle Effect

Customers without additional services like online security or streaming are more likely to discontinue service

Senior Citizen Risk

Senior customers demonstrate higher churn probability, requiring targeted retention strategies

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Classification Models Implemented

Based on the research paper methodology, we implemented supervised learning algorithms taught in class:

Logistic Regression

A statistical model that predicts binary outcomes using a logistic function. It estimates probabilities based on input features and is interpretable for business stakeholders.

Decision Tree

A flowchart-like structure that splits data based on feature values. It's easy to visualize and understand, making it useful for explaining predictions to non-technical audiences.

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Feature Selection Strategy

```
ROUND 1 - ALL FEATURES

Logistic Regression Accuracy: 0.8069552874378992
              precision    recall  f1-score   support

           0       0.85      0.89      0.87      1035
           1       0.66      0.57      0.61       374

    accuracy                           0.81      1409
   macro avg       0.75      0.73      0.74      1409
weighted avg       0.80      0.81      0.80      1409

Decision Tree Accuracy: 0.794180269694819
              precision    recall  f1-score   support

           0       0.84      0.88      0.86      1035
           1       0.63      0.55      0.58       374

    accuracy                           0.79      1409
   macro avg       0.74      0.71      0.72      1409
weighted avg       0.79      0.79      0.79      1409
```

```
ROUND 2 - SELECTED FEATURES

Logistic Regression Accuracy: 0.7977288857345636
              precision    recall  f1-score   support

           0       0.84      0.89      0.87      1035
           1       0.64      0.54      0.59       374

    accuracy                           0.80      1409
   macro avg       0.74      0.72      0.73      1409
weighted avg       0.79      0.80      0.79      1409

Decision Tree Accuracy: 0.794180269694819
              precision    recall  f1-score   support

           0       0.86      0.87      0.86      1035
           1       0.62      0.59      0.60       374

    accuracy                           0.79      1409
   macro avg       0.74      0.73      0.73      1409
weighted avg       0.79      0.79      0.79      1409
```

Round 1: Baseline

Trained models using all available features to establish baseline performance metrics

Round 2: Optimization

Selected important features based on correlation analysis and retrained models

This two-round approach matches the research paper methodology, allowing us to evaluate the impact of feature selection on model performance.

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Model Performance Results

Best Performing Model

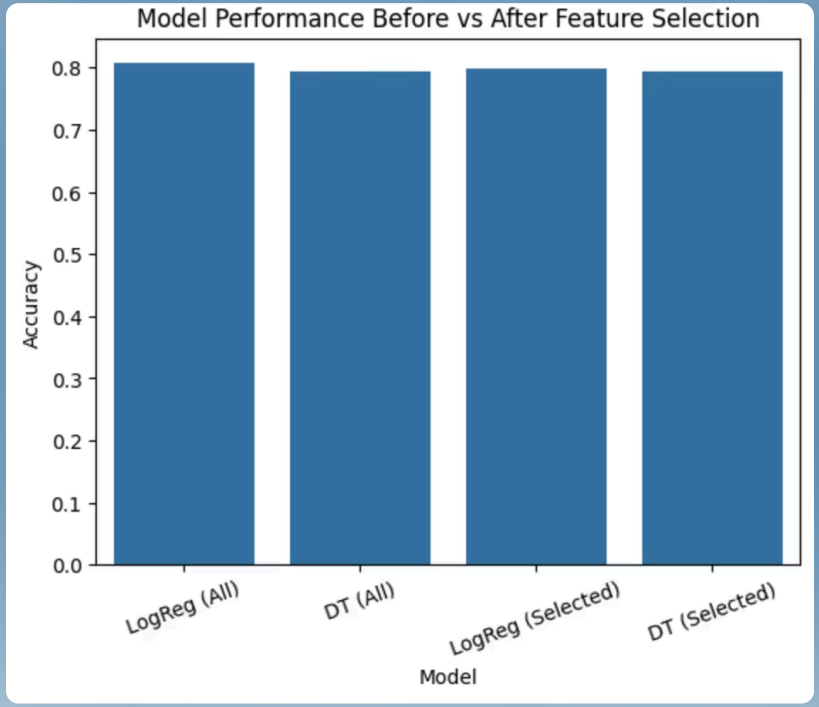
Logistic Regression achieved the highest overall accuracy across both feature selection rounds

Feature Selection Impact

Removing less important features improved model performance slightly while reducing complexity

Model Comparison

Both models performed well, but Logistic Regression showed more consistent results across different metrics



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Implementation: Code Walkthrough

Key Implementation Steps

- **Data Preprocessing**: Cleaned dataset and transformed categorical variables
- **Model Training**: Fitted classification algorithms to training data
- **Model Evaluation**: Calculated accuracy, precision, recall, and confusion matrix

Full implementation is available in my notebook and GitHub.

Data Preprocessing

```
df.drop("customerID", axis=1, inplace=True)

df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df["TotalCharges"].fillna(df["TotalCharges"].median(), inplace=True)

df["Churn"] = df["Churn"].map({"Yes":1, "No":0})
df = pd.get_dummies(df, drop_first=True)
```

Model Training

```
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train_scaled, y_train)

dt = DecisionTreeClassifier(max_depth=5)
dt.fit(X_train, y_train)
```

Model Evaluation

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred_lr))
```

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Potential Model Improvements

Advanced Algorithms

Implement Random Forest or Gradient Boosting for potentially better performance

Hyperparameter Tuning

Use grid search or randomized search to optimize model parameters

Address Class Imbalance

Apply techniques like SMOTE or class weighting to handle unbalanced churn distribution

Feature Engineering

Create new features like customer value segments or usage patterns

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Conclusion & Business Value

Predictive Capability

Machine learning models can effectively predict customer churn with good accuracy

Strategic Decisions

Companies can proactively retain at-risk customers through targeted interventions

Business Impact

Improved customer retention leads to better revenue stability and long-term growth

Customer churn prediction enables proactive retention strategies, helping businesses reduce customer loss and improve profitability.

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Thank You

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